

Tribhuvan University  
Faculty of Humanities & Social Sciences  
OFFICE OF THE DEAN

2021

**Bachelor in Computer Applications**

Course Title: **System Analysis & Design**

Code No: **CACS203** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

**Group B**

2. Describe and compare the various sources of software.
3. What different skills do you need to be a good systems analyst?
4. Describe the steps involved in corporate strategic planning.
5. What are the steps in creating decision table? How do we reduce the size and complexity of a decision table?
6. What is information gathering? Discuss questionnaire along with its merits and demerits.
7. Explain different types of software testing?
8. How can you transform ER Diagram into relation? Explain with your own suitable example.

**Group C**

*Attempt any TWO questions[2x10=20]*

9. Assuming monetary benefits of an information system at Rs 85,000 per year, one-time costs of Rs 75,000, recurring costs of Rs 35,000 per year, a discount rate of 12 percent, and a five-year time horizon, calculate the following:  
a) Net present value of the costs and benefits.  
b) The overall return on investment.  
c) The point at which breakeven occurs.
10. What do you mean by software maintenance? Explain the different types of maintenance? Explain the process of maintaining the information system with examples.
11. Why we draw DFD? Explain the process of drawing DFD using suitable example.